

Wilson (Ke) Zhu

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Permanent Resident in Australia

EDUCATION

Bachelor of Computer Science

February 2022 - December 2025

University of New South Wales

- Distinction academic performance with a **82.9/100 WAM**.
- Participating in the **UNSW ChallEng program, Connected Health project**.
- Recipient of the \$5000 **Taste of Research scholarship** for **Generative AI Teaching Assistant**.
- Recipient of the \$10000 **Honour degree project scholarship** under supervision of **Prof. Ron Van der Meyden**.
- Enrolled in the Computer Science program with **94.95 ATAR for HSC**.

SKILLS/JUNIOR FULL-STACK WEB DEVELOPER

- Programming language: Preferred **Typescript** for frontend and backend web development.
Capable of adopting **Javascript, Java and Python**.
- Technology Stack: Proficient **MongoDB, Express.js, React, Node.js** for full stack project and actively learning **Next.js**.
- Workflow Tool: **Git, CI/CD pipeline, Postman**.
- Testing: Backend unit and integration testing using **Jest** and **SuperTest**.
- Deployment: [Render.com](https://render.com/), **Godaddy**.

RELEVANT SIDE PROJECT/EXPERIENCE

InstaGrade — AI-Powered Short Answer and Essay Grading Platform Jan 2025 - present

Founder & Full Stack Developer | instagrade.com.au

Demo: <https://www.youtube.com/watch?v=MxUhhnTrH-4> (7 mins)

- Designed and implemented the full-stack web application (**UI/UX, API, and AI integration**) with experience of my previous project and assistance of **AI tools** (e.g [Loveable.ai](#) for UI/UX, GPT4o and copilot for identifying bugs).
- Built backend in Node.js and MongoDB database with test coverage; developed frontend fetch, state handling and **responsive** design in **React & Material UI**. Utilizing **Typescript** to ensure type assertion to enhance readability and maintainability of the codebases.
- Integrated GPT API with prompt engineering and RAG to assess and grade answers with tailored feedback.
- Managed **Stripe** subscriptions, free trials, and webhook logic. Enabling users to join and pay monthly subscriptions, check their invoices and update payment details legally and safely without storing sensitive data in my private database.
- Continuously optimized performance, error handling, and database design through receiving feedback from my trial users experience.
- **Deployed to Microsoft Azure**, leveraging Azure Web Apps and DevOps for CI/CD, enabling scalable infrastructure and high availability for concurrent users.

ContriBalance - Assessment Dispute and Feedback System

July 2024 - December 2024

Tech Lead of Team of Five - Backend, Frontend integration & Software Design.

Capstone project - Tech stack : Nextjs, Typescript, Tailwind css, MongoDB, Jest, Azure

- **Suggested the Tech stack**, design the **backend architecture** and **database model**, **setup mongoose to connect MongoDB Atlas**, **setup testing environment** and guidance example to write tests while identifying the strengths of each teammate to **allocate suitable tasks** and promote the team's morale.

- Develop **token** and verification code **based user authentication**, using MongoDB to manage the verification code status and **HTTPS** to protect the sensitive information. Significantly enhance the security level of website application and qualify for deployment.
- Implementing **image upload, delete, storing and fetching using ImageKit**. Enabling students to upload their pdf documents as proof.
- **Designing editable questions form** for course administrators in order to receive expected structured student's dispute. Enabling a more flexible management of the dispute feedback form that administrators can customize.

HeartGPT - Automated ECG Diagnosis Composition System February 2024 - December 2024

Team Member of AI/Transformer training group

University of New South Wales - ChallENG program

- **Participated as a developer** and gained 89/100 final mark and **pitched** our project at the **ChallEng project showcasing event**.
- **Integrated snorkel** to scale the **labeling of raw unlabeled ecg data**. Reduced the time of processing and labelling over 100000 human doctor's diagnoses and building domain knowledge for our Gen AI.
- Implemented functionality to extract **key cardiologist features from ecg data** with AI diagnosis classifier made by previous research then used prompted GPT4 OpenAI api to generate human readable diagnosis, saving doctor's time from writing cardiology report from scratch to validate and minor adjust the generated report.

RELEVANT WORK EXPERIENCE

Student Researcher

January 2024 - May 2024

University of New South Wales - Taste of Research

- Constructed and researched applications of Large Language Models (LLMs) in education.
- Created a model to preprocess student short answer questions and marking criteria into a specific prompt structure.
- Utilized the OpenAI GPT API to accurately assess answers and generate relevant feedback.

Web Developer

August 2023 - January 2024

Wise Stay

- Created and maintained website for Wise Stay, a startup Airbnb hosting company.
- Utilized JavaScript to connect with Airtable for storing property information.
- Developed various sorting options for users.
- Collaborated on backend integrations to ensure smooth payment processing using Stripe and listing management.
- Website url: <https://www.wisestay.com.au>

High School Mathematics Tutor

May 2022 - September 2022

Talent 100

- Taught high school mathematics, planned teaching content, graded homework, and maintained student engagement.

AWARDS

Honour Roll Student - Cumberland High School

Web Application Security- Schools Cyber Security Challenges

Taste of Research - Generative AI Teaching Assistant scholarship